

Simulation Assumption Dependency Module (SADM)

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Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

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Governed Space: Simulation Assumption Dependency

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Canonical Language: English (UCL™)

Governed Space

Simulation Assumption Dependency

Canonical Definition

Simulation Assumption Dependency Module (SADM) defines the governance framework for identifying, structuring, mapping, classifying, propagating, and maintaining the dependencies through which Simulation Assumptions influence other assumptions, model elements, parameters, scenarios, execution conditions, outputs, and downstream simulation conclusions.

It establishes the methodological mechanism through which an identified and sufficiently based Simulation Assumption is connected to everything within the simulation that materially depends upon it.

SADM does not determine whether an assumption is universally true, quantify its complete uncertainty, or determine the ultimate validity of simulation outputs.

It governs where an assumption matters, what depends upon it, how dependencies interact, and how changes or failures in an assumption must remain traceable through the simulation.

Operational Role

SADM serves as the third internal governance space of the Simulation Assumption Standard (SAS).

The preceding modules establish:

SAIM — Which assumptions exist and which are materially relevant?

SABM — On what basis is each assumption being used?

SADM asks:

What depends upon each Simulation Assumption, which assumptions depend upon one another, and how can those dependencies remain visible throughout the simulation lifecycle?

The governed progression is:

Governed Simulation Assumption → Dependency Discovery → Dependency Classification → Dependency Mapping → Dependency Materiality → Dependency Interaction → Dependency Propagation → Dependency Change → Dependency Record

The resulting dependency structure becomes an input to subsequent:

Simulation Assumption Uncertainty → Simulation Assumption Limitation

Module Operational Space

SADM governs: assumption dependency identification, assumption-to-assumption dependency, assumption-to-model dependency, assumption-to-parameter dependency, assumption-to-environment dependency, assumption-to-input dependency, assumption-to-scenario dependency, assumption-to-execution dependency, assumption-to-output dependency, assumption-to-conclusion dependency, direct dependencies, indirect dependencies, upstream dependencies, downstream dependencies, conditional dependencies, shared dependencies, chained dependencies, circular dependencies, hidden dependencies, compound dependencies, dependency concentration, dependency criticality, dependency materiality, dependency interaction, dependency propagation, dependency change, dependency failure propagation, dependency reassessment, and complete assumption-dependency traceability.

Module Function

A Simulation Assumption may appear to be a single statement while influencing a substantial part of the simulation.

For example:

Atmospheric density is assumed to remain within Range X.

That assumption may influence:

Aerodynamic Model → Vehicle Dynamics → Energy Consumption → Trajectory →

Mission Outcome

If the assumption changes, every materially dependent element may also change.

SADM prevents assumptions from being governed as isolated statements when their consequences are structurally distributed throughout the simulation.

Its function is therefore to transform:

Assumption

into:

Assumption + Dependency Structure + Affected Simulation Elements + Propagation
Path Simulation Assumption Dependency

A Simulation Assumption Dependency exists where the state, configuration, behavior, interpretation, execution, or output of one simulation element materially depends upon a Simulation Assumption.

Conceptually:

Simulation Assumption

↓

Dependent Element

↓

Affected Simulation Behavior

A dependency may be explicit or implicit.

SADM requires material dependencies to become governable regardless of how they originated.

Dependency Object

A Simulation Assumption may affect: another assumption, a model component, a simulation environment, a parameter, an input, a transformation, a scenario, an execution condition, an interaction, an output, an interpretation, or a simulation conclusion.

Each materially affected element may become a Dependency Object for purposes of assumption governance.

Direct Dependency

A Direct Dependency exists where an element immediately relies upon an assumption.

Example:

Assumption A → Parameter P

If Parameter P is directly assigned from Assumption A, the dependency is direct.

Indirect Dependency

An Indirect Dependency exists where the influence of an assumption passes through one or more intermediate elements.

Example:

Assumption A → Parameter P → Model M → Output O

Output O may therefore depend materially upon Assumption A even though the output does not reference the assumption directly.

SADM requires material indirect dependencies to remain discoverable.

Upstream Dependency

An assumption may itself depend upon another assumption or governed simulation condition.

Example:

Assumption B depends upon Assumption A.

Assumption A is therefore upstream of Assumption B.

Changes in upstream assumptions may require reassessment of downstream assumptions.

Downstream Dependency

Every materially affected element following an assumption in the dependency chain is downstream of that assumption.

A single assumption may have many downstream dependencies.

SADM preserves these relationships so that assumption change can be propagated rather than treated locally.

Assumption-to-Assumption Dependency

Simulation Assumptions may depend upon one another.

For example:

Assumption A: Road friction remains above threshold X.

Assumption B: Vehicle braking distance remains below threshold Y.

If B relies upon A, the relationship should be explicit.

The simulation should not represent B as independently established where its operation materially depends upon A.

Assumption-to-Model Dependency

A model component may depend upon one or several assumptions.

Example:

Human Response Model

may depend upon assumptions concerning: reaction time, attention, movement speed, compliance, environmental visibility, or behavioral variability.

SADM maps these relationships.

Assumption-to-Parameter Dependency

Parameters frequently encode assumptions.

A fixed parameter may appear to be a technical value while actually representing an assumption about reality.

SADM therefore asks:

Does this parameter derive from an assumption, and would the parameter require reassessment if that assumption changed?

Where yes, the dependency should remain explicit.

Assumption-to-Environment Dependency

A simulation environment may rely upon assumptions concerning: temperature, gravity, terrain, traffic, radiation, network conditions, human presence, resource availability, or other environmental states.

SADM maps the relationship between those assumptions and the affected environmental representation.

Assumption-to-Input Dependency

An input may depend upon assumptions used during: selection, transformation, interpolation, extrapolation, normalization, generation, estimation, or preprocessing.

The input should not be treated as assumption-independent where such dependence materially exists.

Assumption-to-Scenario Dependency

Scenarios may exist only because certain assumptions hold.

For example:

Emergency evacuation completes within five minutes

may be a scenario condition dependent upon assumptions about: population mobility, route availability, communication, human response, and environmental conditions.

SADM makes those dependencies visible.

Assumption-to-Execution Dependency

Simulation execution itself may depend upon assumptions.

Examples include assumptions about: convergence, timing, resource availability, computational behavior, event ordering, system stability, or execution constraints.

Where execution behavior materially depends upon an assumption, the dependency belongs within the governed map.

Assumption-to-Output Dependency

A simulation output may materially depend upon one or many assumptions.

SADM establishes the traceability relationship:

Assumption → Simulation Path → Output

This relationship is critical because a simulation output may appear precise even when it depends heavily upon assumptions.

Assumption-to-Conclusion Dependency

A simulation conclusion may depend upon outputs that themselves depend upon assumptions.

The complete chain may therefore become:

Assumption → Model Element → Execution → Output → Simulation Conclusion

SADM preserves this dependency path.

It does not determine whether the conclusion may legitimately support external reliance; that remains outside the module boundary.

Conditional Dependency

A dependency may exist only under specified conditions.

Example:

Assumption A affects Output O only when Temperature > T.

SADM should preserve the triggering condition rather than treating the dependency as universally active.

Shared Dependency

Multiple simulation elements may depend upon the same assumption.

Conceptually:

Assumption A

↙ ↓ ↘

Element 1 · Element 2 · Element 3

Shared dependencies can create concentration risk because one assumption may influence many apparently independent outputs.

Dependency Concentration

Dependency Concentration exists where a large or critical portion of the simulation depends upon one assumption or a small set of assumptions.

This condition should remain visible.

A simulation containing hundreds of parameters may still depend materially upon only a few foundational assumptions.

SADM prevents structural complexity from hiding assumption concentration.

Compound Dependency

A simulation element may depend simultaneously upon multiple assumptions.

For example:

Output O = dependent upon A + B + C

Where the assumptions interact materially, the dependency should be represented as compound rather than as three unrelated relationships.

Dependency Chain

A Simulation Assumption Dependency Chain represents the sequence through which assumption influence propagates.

Example:

Assumption A

↓

Parameter B

↓

Model Component C

↓

Scenario D

↓

Output E

↓

Conclusion F

SADM requires material chains to remain reconstructable.

Dependency Depth

Dependency Depth describes how far an assumption's influence extends through the simulation structure.

A shallow dependency may affect one parameter.

A deep dependency may propagate through multiple model components, scenarios, outputs, and conclusions.

Depth may therefore be relevant when determining assumption materiality.

Dependency Breadth

Dependency Breadth describes how many materially distinct simulation elements depend upon an assumption.

An assumption may have:

Low Breadth

or:

High Breadth

depending upon how widely its influence is distributed.

Dependency Materiality

Not every dependency requires equal governance attention.

SADM may evaluate materiality according to factors such as: effect on model behavior, effect on scenario behavior, effect on outputs, effect on conclusions, dependency breadth, dependency depth, sensitivity, reversibility, and consequence of dependency failure.

The objective is not to eliminate minor dependencies.

The objective is to ensure that material dependencies cannot remain invisible.

Critical Assumption Dependency

A Critical Assumption Dependency exists where failure, invalidation, or material change of an assumption may substantially alter: simulation behavior, a critical scenario, a key output, a major conclusion, or the intended use of the simulation.

Critical dependencies should receive explicit identification.

Dependency Interaction

Dependencies may interact.

For example:

Assumption A alone may have limited influence.

Assumption B alone may have limited influence.

But:

A + B

may materially change the simulation outcome.

SADM therefore recognizes that dependency governance cannot always be reduced to isolated one-to-one relationships.

Dependency Amplification

A small change in an assumption may produce a disproportionately large downstream effect.

This creates Dependency Amplification.

Example:

Small change in friction assumption

↓

Large change in stopping-distance output

Where amplification is material, the dependency should be identifiable for subsequent uncertainty governance.

Dependency Dampening

The opposite may also occur.

A substantial assumption change may produce little downstream effect because other model mechanisms dampen the influence.

SADM may preserve this relationship where relevant.

Hidden Dependency

A Hidden Simulation Assumption Dependency exists where a material relationship influences the simulation but is not represented within the governed dependency structure.

Hidden dependencies may arise through: inherited models, reused code, default parameters, embedded constants, external libraries, external datasets, model coupling, undocumented transformations, or inherited assumptions.

SADM requires material hidden dependencies, once discovered, to be brought into governance.

Inherited Dependency

A reused model or simulation component may carry dependency relationships from an earlier simulation.

These dependencies should not automatically be assumed applicable.

SADM asks whether they remain relevant to the current: Simulation Object, purpose, scope, reality representation, assumptions, model, environment, and scenario context.

Circular Dependency

A Circular Assumption Dependency exists where assumptions or simulation elements depend recursively upon one another.

Example:

Assumption A → Model B → Derived Condition C → Assumption A

Circularity is not automatically prohibited.

Some dynamic systems legitimately contain feedback loops.

However, material circular dependencies must be visible so they are not mistaken for independent support. Dependency vs. Basis

SADM maintains a strict distinction:

SABM asks: Why is this assumption being used?

SADM asks: What depends upon this assumption?

The basis is upstream justification.

The dependency structure is downstream influence.

The two should not be conflated.

Dependency vs. Uncertainty

SADM identifies the pathways through which an assumption matters.

SAUM — Simulation Assumption Uncertainty Module subsequently governs uncertainty associated with assumptions and its relevance.

SADM therefore identifies:

Where uncertainty may propagate

without replacing uncertainty governance itself. Dependency vs. Limitation

SADM identifies affected simulation elements.

SALM — Simulation Assumption Limitation Module governs where assumptions cease to remain applicable and how those limitations must remain visible.

This preserves clean module boundaries.

Dependency Propagation

Where an assumption changes materially, SADM identifies the downstream path requiring reassessment.

Conceptually:

Assumption Change

↓

Direct Dependencies

↓

Indirect Dependencies

↓

Affected Outputs

↓

Affected Conclusions

This is Simulation Assumption Dependency Propagation.

Dependency Change Trigger

Dependency reassessment may be triggered by: assumption modification, assumption removal, assumption replacement, basis change, model change, environment change, parameter change, scenario change, new dependency

discovery, or changed simulation purpose or scope.

Assumption Removal

Removing an assumption does not necessarily remove its effects.

A parameter, model component, or scenario may continue to encode the old assumption.

SADM therefore requires the affected dependency path to be examined when an assumption is removed.

Assumption Replacement

Where:

Assumption A

is replaced by:

Assumption B

the simulation should not merely update the assumption register.

Material dependencies should be reassessed against the replacement assumption.

Dependency Failure Propagation

Where an assumption becomes unsupported, materially changed, or otherwise unsuitable for its simulation role, dependent elements may require reassessment.

SADM establishes:

Assumption Failure ≠ Local Failure Only

The impact may propagate throughout the simulation.

No Dependency Isolation Principle

SIMULOS® establishes:

A material assumption must not be treated as isolated when simulation elements, outputs, or conclusions materially depend upon it.

No Hidden Propagation Principle

A change to an assumption should not be considered fully governed while material downstream consequences remain unidentified.

No False Independence Principle

SIMULOS® establishes:

Two simulation outputs must not be represented as assumption-independent merely because they are produced by different model components when both materially depend upon the same underlying assumption.

This is particularly important when multiple outputs appear to provide independent confirmation.

No Dependency Laundering Principle

An assumption dependency does not disappear merely because the assumption passes through: a parameter, another model, a scenario, an external component, or multiple computational transformations.

The dependency remains methodologically relevant while material influence persists.

Dependency Traceability

SADM should preserve the relationship:

Assumption

↓

Direct Dependency

↓

Intermediate Dependency

↓

Affected Simulation Element

↓

Affected Output

↓

Affected Conclusion

where applicable.

This creates the Simulation Assumption Dependency Traceability Chain.

Dependency Map

A Simulation Assumption Dependency Map may represent: assumptions, dependency objects, direct relationships, indirect relationships, conditional relationships, shared dependencies, critical dependencies, dependency chains, and affected outputs.

The map may be implemented graphically, computationally, or structurally.

SIMULOS® governs the required methodological relationships rather than prescribing the implementation technology.

Dependency Matrix

Where useful, assumptions and affected elements may be represented through a dependency matrix.

Conceptually: Assumption Model Parameter Scenario Output Conclusion A1 ✓ ✓ — ✓
✓ A2 — ✓ ✓ ✓ — A3 ✓ — ✓ ✓ ✓

The matrix does not replace the underlying dependency records.

It provides a compact representation of them.

Dependency Status

A dependency may be classified as:

Identified

Confirmed

Conditional

Indirect

Critical

Under Assessment

Changed

Inactive

Superseded

or:

Undetermined Dependency Gap

A Simulation Assumption Dependency Gap exists where a material dependency: is unknown, cannot be traced, is incompletely mapped, has ambiguous direction, has uncertain affected elements, or cannot be connected to relevant outputs or conclusions.

Dependency gaps should remain visible.

Dependency Reassessment

When a material assumption or dependency changes, affected relationships should be reassessed.

The reassessment asks:

Does the dependency still exist?

Has its strength or materiality changed?

Have new dependencies appeared?

Have prior dependencies disappeared?

Which downstream elements require renewed examination?

Dependency Continuity

SADM preserves dependency history across simulation changes.

Where assumptions or model configurations change, prior dependency maps should remain traceable to the simulation versions in which they applied.

This supports reconstruction of historical simulation results.

Simulation Version Binding

Dependency structures should be bound to the relevant simulation configuration or version.

A dependency map from Simulation Version 1 should not automatically be treated as valid for Simulation Version 2 where assumptions, models, parameters, or environments materially changed.

Dependency Record

For each material dependency, SADM may establish a Simulation Assumption Dependency Record containing: Dependency ID, Assumption ID, Dependency Object ID, Dependency Type, Dependency Direction, Direct / Indirect Status, Applicable Conditions, Dependency Chain, Dependency Breadth, Dependency Depth, Materiality, Criticality, Interaction References, Propagation Path, Change Status, affected outputs, affected conclusions, simulation version, and traceability references.

Dependency Set

All material assumption dependencies may form a:

Governed Simulation Assumption Dependency Set

This provides a structured basis for subsequent uncertainty and limitation governance.

Minimum Implementation Framework

1. Bind the Dependency Inquiry

Identify the Simulation Assumption and simulation configuration being examined. 2. Identify Direct Dependencies

Determine which assumptions, parameters, models, scenarios, environments, execution conditions, or outputs immediately depend upon the assumption. 3. Trace Indirect Dependencies

Follow material dependency chains through intermediate simulation elements. 4. Classify Dependency Relationships

Distinguish direct, indirect, conditional, shared, compound, inherited, circular, and other relevant dependency forms. 5. Assess Materiality and Criticality

Determine which dependencies can materially influence simulation behavior, outputs, or conclusions. 6. Identify Dependency Interactions

Determine whether multiple assumptions or dependency paths interact, amplify, dampen, or concentrate influence. 7. Establish Propagation Paths

Map how assumption change or failure may affect downstream simulation elements. 8. Record Dependency Gaps

Preserve unknown, incomplete, or unresolved material dependency relationships. 9. Bind Dependencies to Simulation Version

Ensure the dependency structure remains associated with the simulation configuration to which it applies. 10. Preserve Dependency Traceability

Maintain the reconstructable relationship:

**Assumption → Dependency → Simulation
Element → Output → Conclusion**

Governance Outputs

SADM may produce: Simulation Assumption Dependency Identifier, Simulation Assumption Dependency Record, Direct Dependency Record, Indirect Dependency Record, Conditional Dependency Record, Shared Dependency Record, Compound Dependency Record, Circular Dependency Record, Inherited Dependency Record, Critical Dependency Record, Assumption-to-Assumption Dependency Record, Assumption-to-Model Dependency Record, Assumption-to-Parameter Dependency Record, Assumption-to-Scenario Dependency Record, Assumption-to-Execution Dependency Record, Assumption-to-Output Dependency Record, Assumption-to-Conclusion Dependency Record, Dependency Materiality Assessment, Dependency Criticality Assessment, Dependency Interaction Record, Dependency Concentration Record, Dependency Amplification Record, Dependency Propagation Path, Dependency Change Record, Dependency Reassessment Record, Dependency Gap Record, Simulation Assumption Dependency Map, Simulation Assumption Dependency Matrix, Simulation Assumption Dependency Traceability Chain, and Governed Simulation Assumption Dependency Set.

Use Case 1 — Autonomous Vehicle Simulation

Scenario

A vehicle simulation assumes:

Road friction remains above coefficient X.

The assumption is explicitly identified and has a documented engineering basis. Application

SADM maps the dependency chain:

Road-Friction Assumption

↓

Tire-Road Interaction Model

↓

Braking Model

↓

Stopping Distance

↓

Collision-Avoidance Scenario

↓

Safety-Relevant Simulation Output

The same assumption also affects steering stability. Result

The assumption is classified as having:

High Dependency Breadth + Critical Downstream Dependency

A change to the friction assumption therefore requires reassessment of more than the braking model alone. Use Case 2 — Space Mission Simulation

Scenario

A mission simulation assumes:

Solar-array degradation remains within the modeled annual range. Application

SADM identifies dependencies across:

Energy Generation → Battery State → Communications Availability → Thermal Control → Mission Operations

The dependency is conditional upon the simulated radiation environment. Result

The assumption is recorded as a:

Conditional Critical Dependency

If the radiation environment changes materially, the dependency chain becomes subject to reassessment before affected mission outputs are treated as unchanged.

Architectural Position

Simulation Assumption Dependency is the third internal governance space of the Simulation Assumption Standard (SAS).

The complete SAS progression is:

Simulation Assumption Identification → Simulation Assumption Basis → Simulation Assumption Dependency → Simulation Assumption Uncertainty → Simulation Assumption Limitation

SAIM establishes:

Which assumptions exist?

SABM establishes:

Why is each assumption being used?

SADM establishes:

What depends upon each assumption?

SAUM establishes:

What uncertainty exists around the assumptions and how must it remain visible?

SALM establishes:

Where do the assumptions cease to remain applicable and what limitations follow?

Together, these modules ensure that Simulation Assumptions remain identifiable, supported, structurally connected to their consequences, uncertainty-aware, and bounded throughout the simulation lifecycle.

Foundational Principle

A Simulation Assumption cannot be fully governed unless the material simulation elements, outputs, and conclusions that depend upon it can be identified and traced.

Canonical Closing Statement

Simulation Assumption Dependency Module establishes the dependency-governance layer of the Simulation Assumption Standard by identifying and preserving the pathways through which Simulation Assumptions influence other assumptions, parameters, models, environments, scenarios, execution conditions, outputs, and simulation conclusions. Through direct and indirect dependency mapping, shared and compound dependency governance, materiality and criticality assessment, interaction analysis, dependency propagation, change reassessment, version binding, gap visibility, and complete assumption-to-consequence traceability, SADM prevents material assumptions from functioning as isolated or hidden premises and ensures that changes to an assumption can be systematically propagated to every materially affected part of the simulation.

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